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Bereken in $\mathbb{C}$ door gebruik van goniometrische vormen: $\left(\frac{-3-5i}{4+i}\right)^5$

$$\frac{-3-5i}{4+i} = \frac{(-3-5i)(4-i)}{(4+i)(4-i)} = \frac{-12+3i-20i-5}{16+1} = \frac{-17-17i}{17} = -1-i$$

$$z = -1 - i$$

$$r = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}$$

$$\theta = \pi + \text{Bgtan}\left(\frac{-1}{-1}\right) = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$$

$$z = \sqrt{2} \text{cis}\left(\frac{5\pi}{4}\right)$$

$$z^5 = (\sqrt{2})^5 \text{cis}\left(5 \cdot \frac{5\pi}{4}\right) = 4\sqrt{2} \text{cis}\left(\frac{25\pi}{4}\right)$$

$$z^5 = 4\sqrt{2} \text{cis}\left(\frac{\pi}{4}\right)$$

$$z^5 = 4\sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right) = 4\sqrt{2} \left(\frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2}\right)$$

$$z^5 = 4 + 4i$$

References